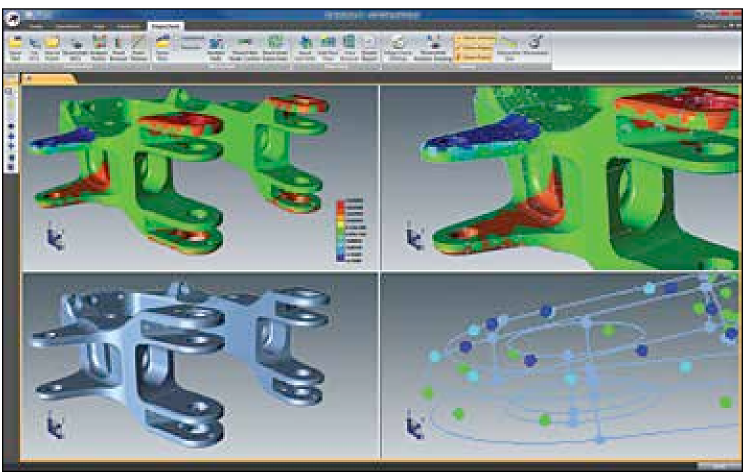
From inception to actualisation

The 3D printing process of any printer can be simplified into a series of basic steps. These steps are independent of the printer's size, scale, material or design, and are closely adhered to by nearly all printer manufacturers.

Step 1: Just as any 2D digital printing begins as a file in a word processing software or page layout software, 3D printing begins in computer-aided design (CAD) software. The version or degree of the software's complexity may vary but they all share the same basic attribute of being able to design a three-dimensional object inside the computer's memory. Depending on the type of software, users can exert various degrees of control over the physical and structural integrity of the final product within the simulated environment of the computer. Data relevant to the product's real world attributes, such as material's property or mensuration can also be accurately depicted using computer-aided design software. The scientific data available to these software is also instrumental in giving users an accurate virtual prototype of their design, allowing them to test how the conceptualised object will behave under a variety of real world settings.

Another highly useful means of obtaining a virtually simulated design of an object is through the use of a 3D scanner. A 3D scanner will allow you to virtually "copy" a physical real world object into a computer by collecting detailed data regarding the object's size, scale, design and composition. This collected data is then exported into CAD software where digital 3D models



The complexity of any product is limited only by your imagination and CAD skills